SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2– Case Study (2 QUESTIONS)

Course Code:	ITA05	**Course Title:**	COMPUTER VISION
CO Assessed:	CO3 – Precision (BL4)	**Assessment:**	Assessment Tool 1 – Case Study
Weightage:	40% of CIA	**Total Marks:**	20 (2 Questions × 10Marks)
CO1 Statement:	*Analyze and extract image features using detection and matching techniques for effective image representation and comparison..(BL4)*		
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Section / Batch:	B.TECH AIML	**Date:**	

2

Code of Conduct

I Logesh.S 192525023 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: logesh S

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Feature Detection and Matching	Preprocessing, Edge Detection, Harris Corner Detection, Hough Transform, SIFT	1	10
Feature Detection and Matching	PCA, Image Representation, Feature Matching, Image Processing Applications	2	10
	GRAND TOTAL:	20 Marks	

Annexure A –Case Study

1. **A remote sensing application is used to align satellite images of the same geographical region captured at different times for monitoring environmental changes. The system initially relies on edge detection to identify matching regions between the images; however, due to differences in illumination, image orientation, and background details, the matching process becomes unreliable and produces inaccurate alignment results. The development team decides to replace the edge-based approach with Harris Corner Detection to identify stable and distinctive feature points. Analyze the limitations of edge detection in this application and explain how Harris Corner Detection improves feature localization, matching accuracy, and robustness for image registration under varying imaging conditions.**

2. **An autonomous surveillance system continuously captures high-resolution color images to detect vehicles and pedestrians in real time. Processing the original color images directly increases computational complexity and slows down feature extraction, making it difficult to achieve the required processing speed. To improve system performance, the engineers consider converting the images into grayscale or binary representations before applying feature detection algorithms. Analyze the impact of different image representations on computational efficiency, edge detection, and feature extraction performance, and justify the most appropriate image representation for achieving accurate and reliable object detection in real-time computer vision applications.**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Faculty In-charge
Signature: ________________
Date: ____________________

ANSWERS

1.Edge detection and Harris Corner Detection are both feature extraction techniques, but they serve different purposes. In remote sensing image registration, Harris Corner Detection provides more reliable feature points for matching satellite images captured under different conditions.

Limitations of Edge Detection in Remote Sensing

1. **Sensitivity to Illumination Changes**
   - Edge detectors (such as Sobel, Prewitt, or Canny) rely on intensity gradients.
   - Changes in sunlight, shadows, seasonal effects, or atmospheric conditions alter image intensity, causing edges to appear differently.
   - This results in missing or false edges, making image matching unreliable.
2. **Poor Feature Uniqueness**
   - Edge pixels lie along continuous boundaries and are not highly distinctive.
   - Many points on the same edge look similar, making it difficult to identify the correct correspondence between two satellite images.
3. **Sensitivity to Orientation**
   - If satellite images are rotated or captured from different viewing angles, edge directions change.
   - Matching based only on edges becomes inaccurate because edge orientation varies.
4. **Influence of Background Details**
   - Natural scenes contain many irrelevant edges from vegetation, clouds, rivers, roads, and buildings.
   - These extra edges increase false matches and reduce registration accuracy.
5. **Weak Localization**
   - An edge indicates only a boundary, not a unique point.

   - Small shifts along an edge cannot be accurately determined, leading to alignment errors.

How Harris Corner Detection Improves Image Registration

Harris Corner Detection identifies **corners**, which are points where intensity changes significantly in multiple directions.

1. Better Feature Localization

- Corners are well-defined, localized points.
- Unlike edges, a corner has a unique position that can be accurately detected.
- This leads to more precise estimation of image alignment.

Example: Building corners, road intersections, and bridge ends remain fixed over time and serve as reliable landmarks.

2. Improved Matching Accuracy

- Corner points are highly distinctive.
- Each corner has a unique neighborhood, making it easier to find the correct matching point in another image.
- This reduces false correspondences compared to edge-based matching.

3. Robustness to Illumination Changes

- Harris Corner Detection uses local intensity variation rather than absolute pixel values.
- Stable corners remain detectable even when lighting conditions, shadows, or seasonal brightness change.
- Therefore, matching remains reliable across images captured at different times.

4. Robustness to Image Rotation

- Harris corners remain detectable even if the image is rotated.
- Although the corner orientation changes, the detector still identifies the same physical feature.

- This improves registration when satellite images have different orientations.

5. Reduced Effect of Background Noise

- Harris focuses on strong corner structures instead of all image boundaries.
- Random texture and weak edges are less likely to be selected.
- This produces a smaller set of reliable feature points, reducing incorrect matches.

6. Higher Registration Accuracy

- Matching several stable corner points allows accurate estimation of transformations such as translation, rotation, and scaling.
- Consequently, satellite images align more precisely for environmental monitoring and change detection.

Comparison of Edge Detection and Harris Corner Detection

Aspect	**Edge Detection**	**Harris Corner Detection**
Feature Type	Boundaries (edges)	Distinctive corner points
Localization	Less precise	Highly precise
Feature Uniqueness	Low	High
Illumination Robustness	Poor	Better
Rotation Robustness	Limited	Good
False Matches	High	Low
Registration Accuracy	Lower	Higher
Suitability for Remote Sensing	Moderate	Excellent

Conclusion

Edge detection is useful for identifying object boundaries but is not ideal for satellite image registration because it is sensitive to illumination changes, orientation differences, and background clutter, and it provides non-unique features that lead to incorrect matches. Harris Corner Detection overcomes these limitations by detecting stable, distinctive, and accurately localized corner points that are more robust to variations in lighting and image orientation. As a result, it significantly improves feature localization, matching accuracy, and the overall robustness of remote sensing image registration for environmental monitoring.

2. Impact of Image Representation on Real-Time Object Detection

In an autonomous surveillance system, the choice of image representation directly affects **processing speed, edge detection, and feature extraction**. Color, grayscale, and binary images have different advantages.

1. Color Image

- Contains three channels: **Red, Green, and Blue (RGB)**.
- Preserves maximum information about vehicles and pedestrians.
- Provides better color-based object separation.
- However, processing three channels requires more memory and computational power.
- Feature extraction and edge detection are slower compared with grayscale.

Limitation: High computational complexity makes direct color processing less suitable for strict real-time applications.

2. Grayscale Image

- Converts the RGB image into a **single intensity channel**.
- Reduces data size and computational requirements significantly.
- Edge detection algorithms such as **Sobel and Canny** work effectively on grayscale images.
- Important shapes, boundaries, textures, and structural features are mostly preserved.
- Faster feature extraction makes it suitable for real-time surveillance.

Advantage: Provides a good balance between **processing speed and useful visual information**.

3. Binary Image

- Represents pixels using only **two values**, usually 0 and 1.
- Requires very little memory and allows extremely fast processing.
- Can clearly separate objects from the background when thresholding is effective.
- However, much of the original image information is lost.

- Fine details, textures, and weak edges may disappear.
- Performance can decrease when lighting or background conditions change.

Limitation: Less reliable for complex scenes containing vehicles and pedestrians at different distances and lighting conditions.

Comparison:

Feature	**Color**	**Grayscale**	**Binary**
Number of channels	3	1	1
Processing speed	Slow	Fast	Very fast
Memory requirement	High	Low	Very low
Edge detection	Good	Excellent	Moderate
Feature information	Very high	High	Low
Detail preservation	Excellent	Good	Poor
Real-time suitability	Moderate	**Excellent**	High, but less reliable

Most Appropriate Representation: Grayscale

Grayscale is the best choice for this application because it significantly reduces computational complexity while preserving important structural information such as **edges, shapes, contours, and object boundaries**.

A suitable pipeline is:

Color Image → Grayscale Conversion → Noise Reduction → Edge/Feature Detection → Object Detection

Binary images can be used later for specific segmentation or background-removal tasks, but using them as the primary representation may lose important details required to distinguish pedestrians and vehicles.

Conclusion

For real-time autonomous surveillance, **grayscale representation provides the best balance between accuracy and computational efficiency**. Color images provide more information but require greater processing, while binary images are faster but lose important features. Therefore, grayscale images are generally the most appropriate input for reliable real-time feature extraction and object detection.